

Question

The following chemical processes occur:

1. A carbon dioxide-saturated solution that has dissolved siderite is oxidized by oxygen to produce a reddish-brown precipitate.
2. A triatomic gas can be prepared by mixing CH_4 , NH_3 , O_2 at high temperature in the presence of a Pt/Tr catalyst.
3. ClO_3F undergoes solvolysis in liquid ammonia, producing two 1:1 type salts.
4. Phosphorus pentafluoride reacts with O_2F_2 at low temperature, producing a 1:1 type salt and releasing a gas.
5. 2,4,6-trinitrophenol and excess bleaching powder solution are added to a calcium hydroxide suspension. An oil-like liquid insoluble in water is obtained in the lower layer of the suspension, and the molecular weight of its components is approximately 164.

For the chemical equations involved in the 5 processes, when the coefficients are balanced to the simplest integer ratio, the following statements are made:

1. The sum of the coefficients of the reactants in process 1 is less than the sum of the coefficients of the products.
2. Two of the four numbers, the sum of the coefficients of the reactants and the sum of the coefficients of the products in process 2, and the sum of the coefficients of the reactants and the sum of the coefficients of the products in process 3, are prime numbers.
3. The sum of the coefficients of the products and the sum of the coefficients of the reactants in process 4 are equal.
4. The sum of the coefficients of the products and the sum of the coefficients of the reactants in process 5 are both prime numbers.

The option containing all the correct statements is:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1, 2

G. 1, 3

H. 1, 4

I. 2, 3

J. 2, 4

K. 3, 4

L. 1, 2, 3

M. 1, 2, 4

N. 2, 3, 4

Answer

Correct Answer: **J**

Detailed Explanation

Process 1: The existing form of siderite FeCO_3 dissolving in carbon dioxide-saturated solution should be $\text{Fe}(\text{HCO}_3)_2$, which is oxidized to ferric hydroxide by oxygen. Balance the equation:

CHECKPOINT

1 PTS

The existing form of siderite FeCO_3 dissolving in carbon dioxide-saturated solution should be $\text{Fe}(\text{HCO}_3)_2$



CHECKPOINT

1 PTS



The sum of reactant coefficients and the sum of product coefficients are 7 and 12, respectively. Statement 1 is incorrect.

Process 2: The triatomic gas is composed of elements C, H, N, O, so it should be HCN ; therefore, balance:

CHECKPOINT

1 PTS

The triatomic gas should be HCN



CHECKPOINT

1 PTS



The sum of reactant coefficients and the sum of product coefficients are 7 and 8, respectively.

Process 3: For the solvolysis reaction, first consider the autoionization of the solvent to produce NH_4^+ , NH_2^- . Therefore, the positive ions of the two salts should both be NH_4^+ , and NH_2^- reacts with ClO_3F to undergo substitution, producing $\text{ClO}_3(\text{NH})^-$. Therefore, balance as follows:

CHECKPOINT

1 PTS

The positive ions should both be NH_4^+ , and NH_2^- reacts with ClO_3F to produce $\text{ClO}_3(\text{NH})^-$



CHECKPOINT

1 PTS



The sum of reactant coefficients and the sum of product coefficients are 4 and 2, respectively.

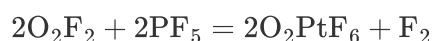
There are two prime numbers in 4, 2, 7, 8. Statement 2 is correct.

Process 4: O_2F_2 is a strong oxidant, and the positive ion can only be O_2^+ ; PF_5 acts as a Lewis acid to produce PF_6^- , and the gas released at the same time should be fluorine gas. Balance as follows:

CHECKPOINT

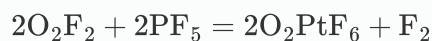
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The positive ion can only be O_2^+ ; PF_5 acts as a Lewis acid to produce PF_6^-



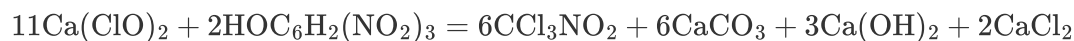
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1 PTS



The sum of reactant coefficients and the sum of product coefficients are 4 and 3, respectively. Statement 3 is incorrect.

Process 5: The main component of bleaching powder is $\text{Ca}(\text{ClO})_2$, in which $\text{Cl}(\text{I})$ has strong oxidizing properties and can oxidize the carbon of organic matter to CO_3^{2-} ; the produced insoluble liquid is obviously an organic chloride. Considering the molecular weight, the liquid component is CCl_3NO_2 . Therefore, balance the equation:



CHECKPOINT

1 PTS

The main component of bleaching powder is $\text{Ca}(\text{ClO})_2$, $\text{Cl}(\text{I})$ can oxidize the carbon of organic matter to CO_3^{2-}

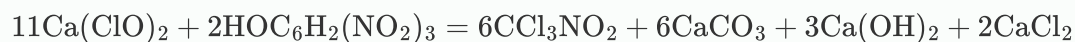
CHECKPOINT

1 PTS

The insoluble liquid is obviously an organic chloride. Considering the molecular weight, the liquid component is CCl_3NO_2

CHECKPOINT

1 PTS



The sum of reactant coefficients and the sum of product coefficients are 13 and 17, respectively. Statement 4 is correct.

Judge the statements based on the equations. Statements 1 and 3 are incorrect, and statements 2 and 4 are correct. Option J is correct.